PCSE – REAL LIFE PROJECT-2

Guidelines for Selecting a Software System / Mobile Application

WHAT IS REAL LIFE PROJECT–2?

- Design and develop a working software system or mobile application
- Solve a real-world problem
- Apply knowledge from multiple subjects
- Demonstrate technical and professional skills
- Note: This is not just a coding task — it is a complete system development experience.

CHARACTERISTICS OF A STRONG PROJECT

A good project should:

- Solve a real problem
- Have clearly defined users
- Be technically feasible
- Demonstrate course learning outcomes
- Include some innovation
- Note: Emphasis should be placed more on practical relevance than complexity.

START WITH A REAL PROBLEM

Identify real issues in:

- College administration
- Local businesses
- Hospitals / Clinics
- NGOs
- Retail shops
- Transportation

START WITH A REAL PROBLEM ….CONTND

Ask:

- Who is facing the problem?
- How is it currently solved?
- What are the limitations?

Example Ideas:

- Attendance Tracking System
- Shop Inventory System
- Appointment Booking App
- Complaint Management System

KNOW YOUR USERS

- Define:
- Who will use your system?
  - Students?
  - Teachers?
  - Shop owners?
  - Customers?
- What is their technical level?
- What devices do they use?
  - Android phones?
  - Web browsers?
  - Desktop systems?

Note: A project without defined users is not a real project.

REAL LIFE PROJECT 2 – CORE OBJECTIVE

- Demonstrate advanced system development skills
- Integrate multiple technologies
- Apply software engineering lifecycle
- Work with minimal supervision

PROJECT FEASIBILITY

Before choosing, check:

- Time available (semester duration)
- Team size
- Available technical skills
- Required tools & software
- Internet or hardware needs

Avoid:

- Large (enterprise) systems
- Complex AI without knowledge
- Projects requiring expensive hardware

SCOPE MANAGEMENT

Define:

- Must-have features
- Nice-to-have features

Focus on:

- Quality over quantity
- Working system over incomplete big system

Example:

- ❌ "Complete E-commerce System like Amazon"
  ✓ "Local store ordering system with cart and admin panel"

TECHNOLOGY STACK SELECTION

Choose technology based on your skills.

- Web Application:
  - HTML, CSS, JavaScript
  - PHP / Node.js / Python
  - MySQL / MongoDB
- Mobile Application:
  - Android (Java/Kotlin)
  - Flutter
  - React Native
- Also consider:
  - Database design
  - Security basics
  - User authentication

INNOVATION & VALUE ADDITION

Your project should include:

- Improved user experience
- Automation of manual tasks
- Reports or dashboard
- Analytics
- Performance improvements
- Note: Add something unique beyond basic CRUD operations

ETHICAL & LEGAL CONSIDERATIONS

Put the following into consideration:

- Protect user data
- Secure password storage
- Avoid plagiarism
- Do not copy full projects
- Respect copyright

TESTING & USABILITY

Check:

- Easy navigation
- No crashes
- Proper input validation
- Tested with real/potential users
- Works on different devices
- Note: Testing is compulsory, not optional

DOCUMENTATION REQUIREMENTS

Your project report must include:

- Problem statement
- Objectives
- System requirements
- ER Diagram
- Use Case Diagram
- System architecture
- Testing report
- Future enhancements

EVALUATION CRITERIA

Example Mark Distribution rubric:

- Problem relevance – 15%
- System design – 15%
- Functionality – 25%
- Database design – 10%
- UI/UX – 10%
- Innovation – 10%
- Documentation – 10%
- Presentation – 5%

SUGGESTED PROJECT DOMAINS

- College Results Management System
- Library Management System
- Hostel Management System
- Inventory System for Small Shop
- Online Complaint Portal
- Bus Tracking App
- Clinic Appointment System
- Event Management System

PROJECTS TO AVOID

- Copied GitHub projects
- Extremely complex systems
- Static websites only
- No-database projects
- Game-only projects (unless approved)

PROJECT APPROVAL CHECKLIST

- As you seek approval, please ensure:
  - ✓ Clear problem statement
  - ✓ Identified users
  - ✓ Defined features
  - ✓ Selected technology
  - ✓ Feasible within time
  - ✓ Approved by guide

IN-CLASS ACTIVITY

Come up with:

- Three real-life problems
- Choose one
- Define:
  - Target users
  - Core features
  - Technology stack
  - Expected outcome

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PROJECT PROPOSAL TEMPLATE

REAL LIFE PROJECT-2
PROJECT PROPOSAL

- **Project Title:**
  (Enter Your Project Name Here)
- **Submitted By:**
- Student Name – Regl No
- **Course:** Certificate in _____
  Subject: Real Life Project – 2
  Guide Name:
  Institution Name:
  Academic Year:

PROBLEM STATEMENT

- Clearly describe:
  - What is the problem?
  - Who is facing this problem?
  - Why is it important to solve?
- (Write 5–8 clear sentences)

EXISTING SYSTEM

- Describe:
  - How is the problem currently solved?
  - What are the limitations?
    - Manual process?
    - Time-consuming?
    - Error-prone?
    - No proper records?

PROPOSED SYSTEM

- Explain:
  - What system/application are you proposing?
  - How will it solve the problem?
  - What makes it better than the existing system?

OBJECTIVES OF THE PROJECT

- List clear objectives:
  - To develop a _____ system
  - To automate _____
  - To reduce _____
  - To improve _____
  - To provide secure _____
- (4–6 measurable objectives)

SCOPE OF THE PROJECT

- Define:
  - ✓ What the system WILL do
  - ✗ What the system will NOT do
- This prevents scope creep.

TARGET USERS

Identify:

- Primary users
- Secondary users (if any)
- Technical level of users
- Devices they will use

Example:

- Admin
- Registered users
- Guest users

FUNCTIONAL REQUIREMENTS

List main features:

- User Registration/Login
- Add / Edit / Delete Records
- Search Function
- Reports Generation
- Admin Dashboard
- Notifications

- (Use bullet points)

NON-FUNCTIONAL REQUIREMENTS

- Specify:
  - ✓Security
  - ✓Performance
  - ✓Usability
  - ✓Reliability
  - ✓Compatibility
  - ✓Data backup

TECHNOLOGY STACK

Specify clearly:

- Frontend:
  - (HTML/CSS/JS/Flutter/etc.)
- Backend:
  - (PHP / Node.js / Python / etc.)
- Database:
  - (MySQL / MongoDB / etc.)
- Platform:
  - Web / Android / Cross-platform

SYSTEM ARCHITECTURE

Include:

- Simple block diagram
- Client → Server → Database structure
- OR Mobile App → API → Database

(Students must draw a diagram here)

DATABASE DESIGN

Include:

- List of main tables
- Key attributes
- Primary key
- Foreign key relationships

(Optional: Insert ER Diagram)

USE CASE DIAGRAM

Include:

- Actors
- Major system interactions

(Students may draw diagram using UML tools)

PROJECT TIMELINE

- Use simple timeline:

Week	Task
1–2	Requirement Analysis
3–4	System Design
5–8	Development
9	Testing
10	Documentation
11	Final Review

FEASIBILITY ANALYSIS

Include:

- Technical feasibility
- Time feasibility
- Economic feasibility
- Operational feasibility

RISK ANALYSIS

- Identify:
  - Lack of time
  - Technical difficulty
  - Data loss
  - Team coordination issues
- Also mention mitigation strategy.

EXPECTED OUTCOMES

- Describe:
  - ✓What will be achieved?
  - ✓Who will benefit?
  - ✓Long-term impact?

FUTURE ENHANCEMENTS

- Mention:
  - ✓ Mobile version
  - ✓ Cloud deployment
  - ✓ Payment integration
  - ✓ AI features
  - ✓ Multi-language support

CONCLUSION

Summarize:

- Problem
- Proposed solution
- Importance
- Feasibility

(5–6 strong closing sentences)